

Q Assignment - 1

Q1) Greedy algorithm for multi-Armed Bandit Problem

given

machine - A [3, 2, 4]

machine - B [6, 7]

machine C [5, 4, 6]

formula $Q(a) = \frac{\text{sum of rewards}}{N(a)}$

$$\text{machine A} = \frac{3 + 2 + 4}{3} = 3.0$$

$$\text{machine B} = \frac{6 + 7}{2} = 6.5$$

$$\text{machine C} = \frac{5 + 4 + 6}{3} = 5.0$$

The Greedy Option is B = 6.5

Q2) Greedy method using incremental method.

Given

machine A [3, 2, 4]

machine B [6, 7]

machine C [5, 4, 6]

$$Q(A) = 0$$

$$N(A) = 0$$

$$Q(B) = 0$$

$$N(B) = 0$$

$$Q(C) = 0$$

$$N(C) = 0$$

formula

$$Q_{\text{new}}(a) = Q_{\text{old}}(a) + \frac{R - Q_{\text{old}}(a)}{N(a)}$$

Machine A:- Rewards = 3, 2, 4

First reward = 3

initially =

$$Q(A) = 0$$

$$N(A) = 0 + 1$$

$$N(A) = 1$$

$$\text{update } Q(A) = 0 + \frac{3-0}{1}$$

$$Q(A) = 3$$

Second $R = 2$

$$N(A) = 2$$

$$Q_{\text{old}}(A) = 3 \quad 0.5 = \frac{1+3+3}{5} = 2 \text{ window}$$

$$Q(A) = 3 + \frac{2-3}{2}$$

$$Q(A) = 3 - \frac{1}{2}$$

$$Q(A) = 2.5$$

Third $R = 4$

$$N(A) = 3$$

$$Q_{\text{old}}(A) = 2.5$$

$$Q_{\text{new}}(A) = 4.5$$

Third $R = 6$

$$N(C) = 3$$

$$Q_{\text{old}} = 4.5$$

$$Q(C) = 4.5 + \frac{6-4.5}{3}$$

Machine B (reward 6, 7)

First $R = 6$

$$N(B) = 1$$

$$Q_{\text{old}}(B) = 0$$

$$Q(B) = 0 + \frac{6-0}{1}$$

$$Q(B) = 6$$

Therefore

$$Q(A) = 3.0$$

$$Q(B) = 6.5$$

$$Q(C) = 5.0$$

Second $R=7$

$$N(B)=2$$

$$Q_{old}(B)=6$$

$$Q(B) = \frac{6+7}{2}$$

$$= 6 + \frac{1}{2}$$

$$Q(B) = 6.5$$

The highest value is

$$Q(B) = 6.5$$

final machine B

$$N(B)=2$$

$$Q(B)=6.5$$

machine C (rewards: 5, 4, 6)

First $R=5$

$$N(C)=1$$

$$Q_{old}(C)=0$$

$$Q_{new}(C)=5$$

Second $R=4$

$$N(C)=2$$

$$Q_{old}(C)=5$$

Greedy method using incremental method

Given machine A [3, 2, 9]
machine B [1, 6, 4]
machine C [5, 4, 6]

Q(A)=0
Q(B)=0
Q(C)=0

$$Q_{new}(A) = Q_{old}(A) + \frac{R - Q_{old}(A)}{N(A)}$$